To make things easier in this book, all of our schematics have had an accompanying illustration to provide a little better sense of the completed prototyped circuit, but once you get a feeling for what all those symbols mean then it's not really all that bad. Being able to read a schematic also makes it a lot easier to wire things up. Some of the more common schematic symbols are shown in Figure 12-11.

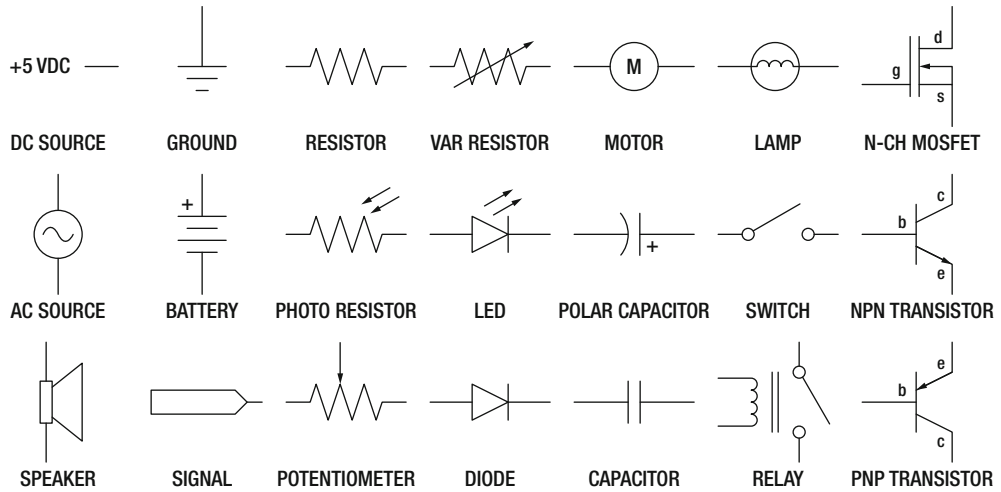

Figure 12-11. *Common schematic symbols*

While these symbols are somewhat abstract, they are also loosely based on reality. Some resistors are made by winding a long length of wire very tightly in a small package, capacitors have two plates that are sometimes separated by air, and diodes let electricity pass in one direction but block it in the other. By reading these symbols, we can use these schematics to build prototype circuits on our breadboards.

Prototyping

Prototyping is a way to quickly realize a circuit and to see if it will work or not. By prototyping a thing, we can build something without spending a lot of time and money wiring everything up from scratch. This is great for focusing our efforts and our ideas rather than getting lost in soldering wires from one component to another. One of the best things that we can use to help make prototyping even easier is a breadboard.

Breadboards

Historically, amateur radio builders would put together circuits using wires and nails quite literally on cutting boards for bread. Today, we use the solderless breadboard, shown in Figure 12-12, to quickly prototype a circuit.